

Depth Anything: Unleashing the Power of Large-Scale Unlabeled Data

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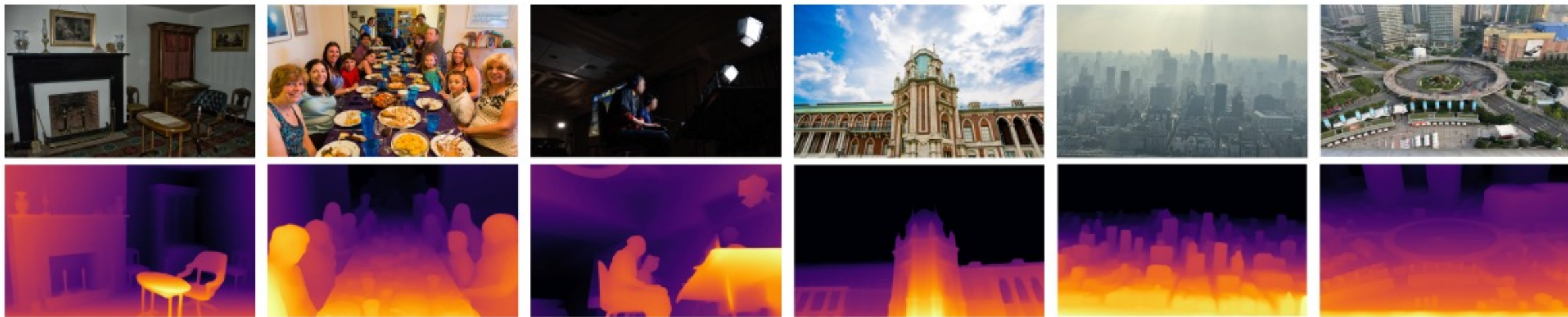
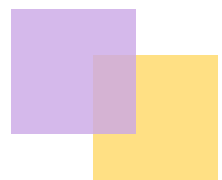


Figure 1. Our model exhibits impressive generalization ability across extensive unseen scenes. **Left two columns:** COCO [36]. **Middle two:** SA-1B [27] (a hold-out unseen set). **Right two:** photos captured by ourselves. Our model works robustly in low-light environments (1st and 3rd column), complex scenes (2nd and 5th column), foggy weather (5th column), and ultra-remote distance (5th and 6th column), *etc.*

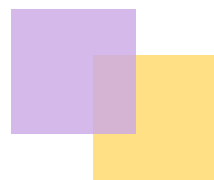
A large, stylized purple number '04' is centered in the background. The '0' is a thick, rounded shape, and the '4' is composed of a thick diagonal stroke and a thick vertical stroke.

Introduction



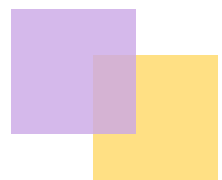
1. Introduction

- The field of computer vision and natural language processing is currently experiencing a revolution with the emergence of “foundation models” that **demonstrate strong zero-/few shot performance in various downstream scenarios.**
→ rely on large-scale training data that can effectively cover the data distribution
- Monocular Depth Estimation(MDE) also requires a foundation model to estimate depth information from a single image.
→ however, this has been underexplored **due to the difficulty of building datasets with tens of millions of depth labels.**



1. Introduction

- In this work, our goal is to build a foundation model for MDE capable of producing high-quality depth information **for any images under any circumstances**.
→ approach this target from the perspective of dataset scaling-up.
- Traditionally, depth datasets are created mainly by acquiring depth data from sensors, stereo matching or SfM, which is **costly, time-consuming, or even intractable in particular situations**.
→ pay attention to large-scale **unlabeled data**



1. Introduction

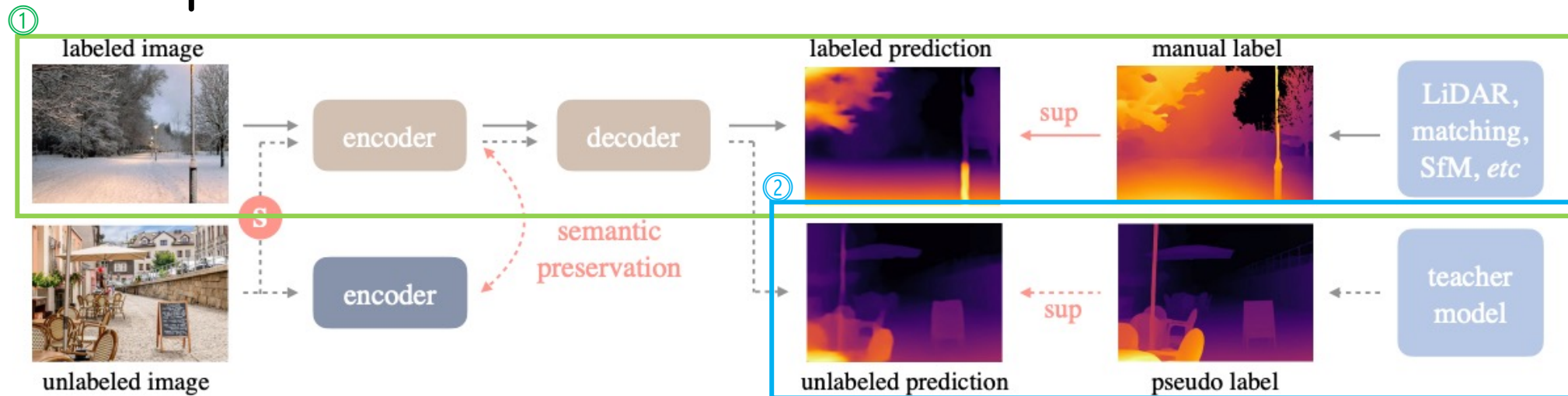
- Compared with stereo images or labeled images from depth sensors, our used monocular unlabeled images exhibit three advantages:
 - (i) **(simple and cheap to acquire)** Monocular images exist almost everywhere, thus they are easy to collect, without requiring specialized devices.
 - (ii) **(diverse)** Monocular images can cover a broader range of scenes, which are critical to the model generalization ability and scalability.
 - (iii) **(easy to annotate)** We can simply use a pre-trained MDE model to assign depth labels for unlabeled images, which only takes a feedforward step. More than efficient, this also produces denser depth maps than LiDAR [18] and omits the computationally intensive stereo matching process.



Method

2. Depth Anything

- Our work utilizes both labeled and unlabeled images to facilitate better monocular depth estimation (MDE).
- Formally, the labeled and unlabeled sets are denoted as $D^l = \{(x_i, d_u)\}_{i=1}^M$ and $D^u = \{u_i\}_{i=1}^N$ respectively.
- Step1 : We aim to learn a teacher model T from D^l .
Step2 : Then, we utilize T to assign pseudo depth labels for D^u .
Step3 : Finally, we train a student model S on the combination of labeled set and pseudo labeled set.





2.1 Learning Labeled Images

- This process is **similar to the training of MiDaS**. However, since MiDaS did not release its code, we first reproduced it.
 - Various datasets containing depth information **are not compatible in terms of scale and shift**.
 - This is due to the diversity of measuring tools, including stereo cameras, laser scanners, and light sensors.
 - : *MidaS* introduces a **new loss function that absorbs these diversities**.
- The depth value is first transformed into the disparity space by $d = 1/t$ and then normalized to 0~1 on each depth map.

affine-invariant loss :

$$\mathcal{L}_l = \frac{1}{HW} \sum_{i=1}^{HW} \rho(d_i^*, d_i),$$



2.1 Learning Labeled Images

- $\rho(d_i^*, d_i) = |\hat{d}_i^* - \hat{d}_i|$, where $\hat{d}_i^*$ is the scaled and shifted versions of the prediction d_i^* and $\hat{d}_i$ is the scaled and shifted versions of the ground truth d_i :

$$\hat{d}_i = \frac{d_i - t(d)}{s(d)},$$

where $t(d)$ and $s(d)$ are used to align the prediction and ground truth to have zero translation and unit scale:

$$t(d) = \text{median}(d), \quad s(d) = \frac{1}{HW} \sum_{i=1}^{HW} |d_i - t(d)|.$$

2.1 Learning Labeled Images

- To obtain a robust monocular depth estimation model, we collect **1.5M labeled images** from 6 public datasets. We use fewer labeled datasets than MiDaS v3.1
- Despite using fewer labeled images, our **easy-to-acquire and diverse unlabeled images** will comprehend the data coverage and greatly enhance the model generalization ability and robustness.

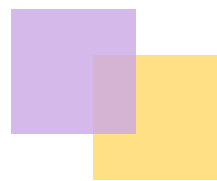
Dataset	Indoor	Outdoor	Label	# Images
Labeled Datasets				
BlendedMVS [77]	✓	✓	Stereo	115K
DIML [13]	✓	✓	Stereo	927K
HRWSI [68]	✓	✓	Stereo	20K
IRS [62]	✓		Stereo	103K
MegaDepth [33]		✓	SfM	128K
TartanAir [63]	✓	✓	Stereo	306K

2.2 Unleashing the Power of Unlabeled Images

- Distinguished from prior works that laboriously construct diverse labeled datasets, we highlight the value of unlabeled images in enhancing the data coverage.
→ We select eight large-scale public datasets as our unlabeled sources for their diverse scenes.
They contain **more than 62M images in total**. The details are provided in the bottom half of Table 1.

Unlabeled Datasets				
BDD100K [82]		✓	None	8.2M
Google Landmarks [65]		✓	None	4.1M
ImageNet-21K [50]	✓	✓	None	13.1M
LSUN [81]	✓		None	9.8M
Objects365 [53]	✓	✓	None	1.7M
Open Images V7 [30]	✓	✓	None	7.8M
Places365 [88]	✓	✓	None	6.5M
SA-1B [27]	✓	✓	None	11.1M

Table 1. In total, our Depth Anything is trained on 1.5M labeled images and **62M unlabeled images** jointly.



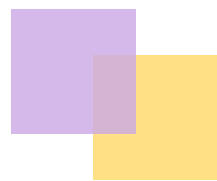
2.2 Unleashing the Power of Unlabeled Images

- Technically, given the previously obtained MDE teacher model T , we make predictions on the unlabeled set D^u to obtain a pseudo labeled set $\hat{D}^u$:

$$\hat{D}^u = \{(u_i, T(u_i)) | u_i \in \mathcal{D}^u\}_{i=1}^N.$$

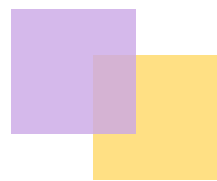
With the combination set $D^l \cup \hat{D}^u$ of labeled images and pseudo labeled images, we train a student model S on it.

- Instead of fine-tuning S from T , **we re-initialize S for better performance.**



2.2 Unleashing the Power of Unlabeled Images

- Unfortunately, in our pilot studies, we failed to gain improvements with such a self-training pipeline, which indeed contradicts the observations when there are only a few labeled images
- We conjecture that, with already sufficient labeled images in our case, the extra knowledge acquired from additional unlabeled images is rather limited.
- Especially considering the teacher and student share the same pre-training and architecture, they **tend to make similar correct or false predictions on the unlabeled set D^u** , even without the explicit self-training procedure.



2.2 Unleashing the Power of Unlabeled Images

- To address the dilemma, we propose to challenge the student with a more difficult optimization target for **additional visual knowledge on unlabeled images**.
 - We inject strong perturbations to unlabeled images during training.
 - : It compels our student model to actively seek extra visual knowledge and acquire invariant representations from these unlabeled images.
 - help our model deal with the open world more robustly.
- Two forms of perturbations:
 - one is strong color distortions, including color jittering and Gaussian blurring
 - the other is strong spatial distortion, which is CutMix

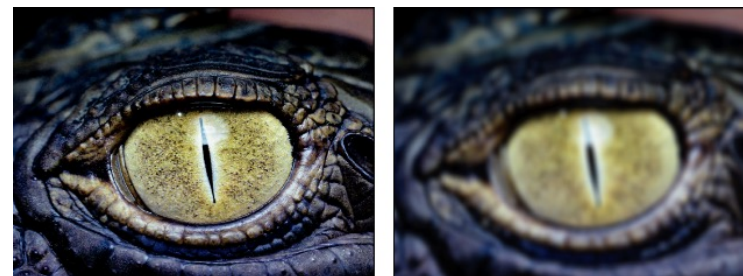
2.2 Unleashing the Power of Unlabeled Images

- Color distortion

Color jittering



Gaussian blurring



- CutMix

	ResNet-50	Mixup [48]	Cutout [3]	CutMix
Image				
Label	Dog 1.0	Dog 0.5 Cat 0.5	Dog 1.0	Dog 0.6 Cat 0.4

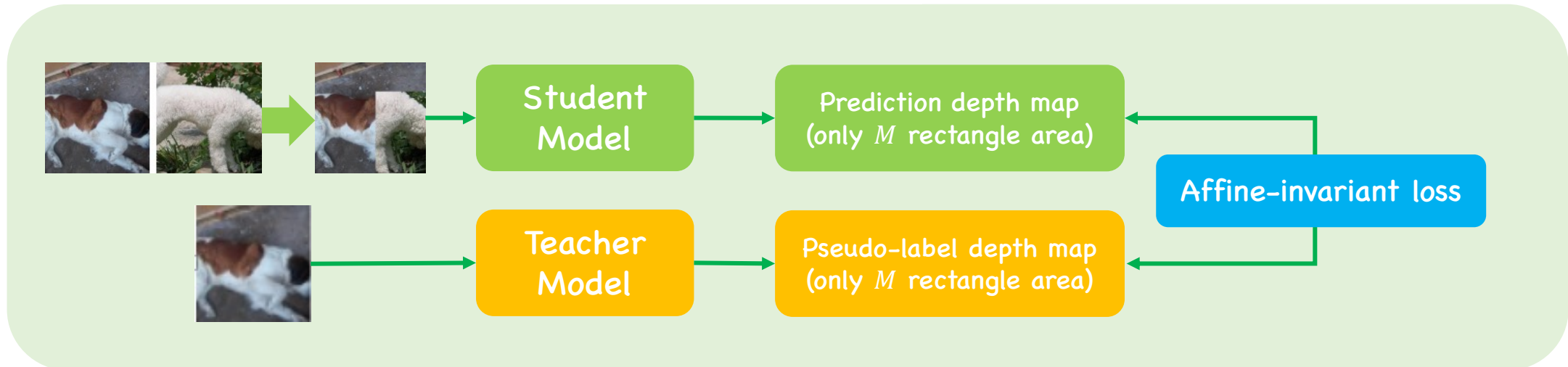
2.2 Unleashing the Power of Unlabeled Images

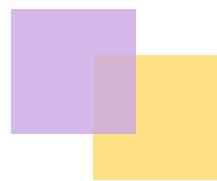
- We first interpolate a random pair of unlabeled images u_a and u_b spatially:

$$u_{ab} = u_a \odot M + u_b \odot (1 - M),$$

- The unlabeled loss $\mathcal{L}_u$ is obtained by first computing affine-invariant losses in valid regions defined by M and $1 - M$, respectively:

$$\begin{aligned} \mathcal{L}_u^M &= \rho(S(u_{ab}) \odot M, T(u_a) \odot M), \\ \mathcal{L}_u^{1-M} &= \rho(S(u_{ab}) \odot (1 - M), T(u_b) \odot (1 - M)), \end{aligned} \rightarrow \rightarrow \mathcal{L}_u = \frac{\sum M}{HW} \mathcal{L}_u^M + \frac{\sum (1 - M)}{HW} \mathcal{L}_u^{1-M}.$$





2.3 Semantic-Assisted Perception

- In our specific context of leveraging unlabeled images, these auxiliary supervision signals from other tasks can also combat the **potential noise in our pseudo depth label**.
→ We aim to seek more informative semantic signals to serve as auxiliary supervision for our depth estimation task.
- We are greatly astonished by the strong performance of DINOv2 models in semantic-related tasks, e.g., image retrieval and semantic segmentation, even with frozen weights without any fine-tuning.
- Motivated by these clues, we propose to transfer its strong semantic capability to our depth model **with an auxiliary feature alignment loss**.

2.3 Semantic-Assisted Perception

- The feature alignment loss is formulated as:

$$\mathcal{L}_{feat} = 1 - \frac{1}{HW} \sum_{i=1}^{HW} \cos(f_i, f'_i),$$

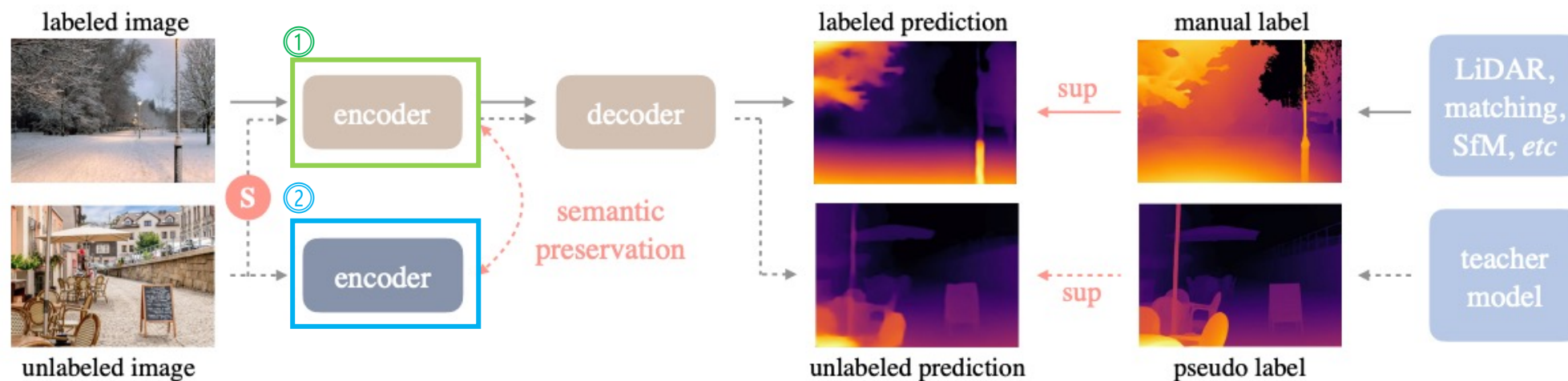
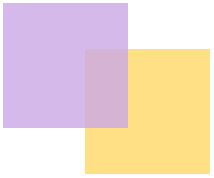


Figure 2. Our pipeline. Solid line: flow of labeled images, dotted line: unlabeled images. We especially highlight the value of large-scale unlabeled images. The **S** denotes adding strong perturbations (Section 3.2). To equip our depth estimation model with rich semantic priors, we enforce an auxiliary constraint between the **online student model** and a **frozen encoder** to preserve the semantic capability (Section 3.3).



2.3 Semantic-Assisted Perception

- To solve this issue, we set a tolerance margin α for the feature alignment. If the cosine similarity of f_i and f'_i has surpassed α , **this pixel will not be considered in our L_{feat} .**
If $\cos(f_i, f'_i) > \alpha$, then L_{feat} is not considered.
- This allows our method to enjoy both the semantic-aware representation from DINOv2 and the part-level discriminative representation from depth supervision.
- As a side effect, our produced encoder not only performs well in downstream MDE datasets, but also achieves **strong results in the semantic segmentation task.**
- Finally, our overall loss is an average combination of the three losses L_l , L_u , and L_{feat} .

Question



Experiment

3.1 Implementation Details

- We adopt the DINOv2 encoder for feature extraction.
- Following MiDaS, we use the DPT decoder for depth regression.

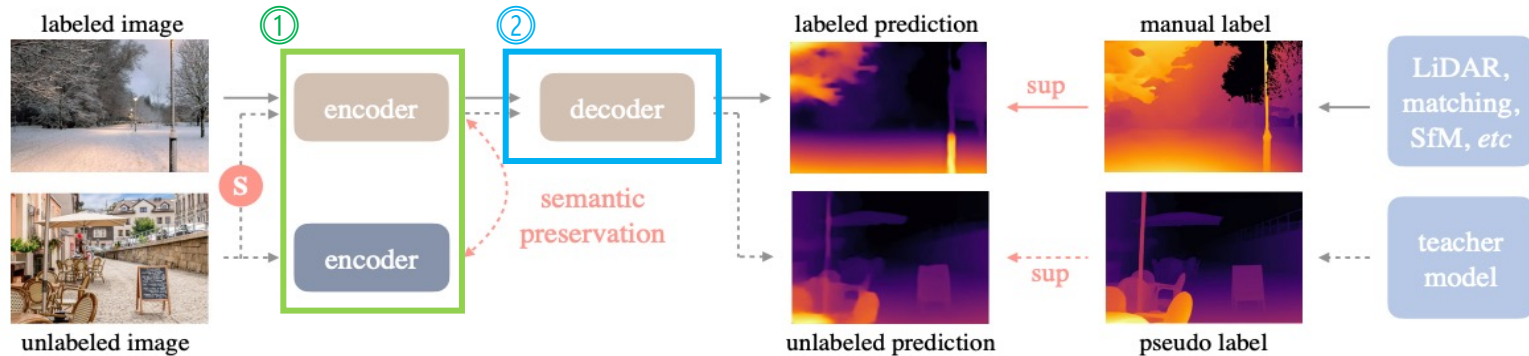


Figure 2. Our pipeline. Solid line: flow of labeled images, dotted line: unlabeled images. We especially highlight the value of large-scale unlabeled images. The **S** denotes adding strong perturbations (Section 3.2). To equip our depth estimation model with rich semantic priors, we enforce an auxiliary constraint between the **online student model** and a **frozen encoder** to preserve the semantic capability (Section 3.3).

- Stage1 : we train a teacher model on labeled images for 20 epochs.
- Stage2 : In joint training, we train a student model to sweep across all unlabeled images for one time.
- The ratio of labeled and unlabeled images is set as 1:2 in each batch.

3.2 Zero-Shot Relative Depth Estimation

- This work aims to provide accurate depth estimation **for any image**. Therefore, we comprehensively validate the **zero-shot depth estimation capability of our Depth Anything model** on six representative unseen datasets: KITTI, NYUv2, Sintel, DDAD, ETH3D, and DIODE.
- We compare with the best DPT-BEiT-L-512 model from the latest MiDaS v3.1, which uses more labeled images than us.

Method	Encoder	KITTI [18]		NYUv2 [55]		Sintel [7]		DDAD [20]		ETH3D [52]		DIODE [60]	
		AbsRel	δ_1	AbsRel	δ_1	AbsRel	δ_1	AbsRel	δ_1	AbsRel	δ_1	AbsRel	δ_1
MiDaS v3.1 [5]	ViT-L	0.127	0.850	0.048	<u>0.980</u>	0.587	0.699	0.251	0.766	0.139	0.867	0.075	0.942
	ViT-S	0.080	0.936	0.053	0.972	0.464	0.739	0.247	0.768	0.127	0.885	0.076	0.939
Depth Anything	ViT-B	0.080	0.939	0.046	0.979	0.432	0.756	0.232	0.786	0.126	0.884	0.069	0.946
	ViT-L	0.076	0.947	0.043	0.981	<u>0.458</u>	0.760	0.230	0.789	<u>0.127</u>	0.882	0.066	0.952

Table 2. **Zero-shot relative** depth estimation. **Better:** AbsRel $\downarrow$, δ_1 $\uparrow$. We compare with the best model from MiDaS v3.1. Note that MiDaS *does not* strictly follow the zero-shot evaluation on KITTI and NYUv2, because it uses their training images. We provide three model scales for different purposes, based on ViT-S (24.8M), ViT-B (97.5M), and ViT-L (335.3M), respectively. **Best**, second best results.



3.3 Fine-tuned to Metric Depth Estimation

- we further examine our Depth Anything model as a promising weight initialization for downstream metric depth estimation.
→Metric Depth Estimation:
: **estimate depth in real-world units such as meters or centimeters.**
- We initialize the encoder of downstream MDE models with our pre-trained encoder parameters and leave the decoder randomly initialized.
- The model is fine-tuned with corresponding metric depth information. In this part, we use our ViT-L encoder for fine-tuning.
- We examine two representative scenarios:
 - 1) in-domain metric depth estimation, where the model is trained and evaluated on the same domain
 - 2) zero-shot metric depth estimation, where the model is trained on one domain, e.g., NYUv2, but evaluated in different domains, e.g., SUN RGB-D

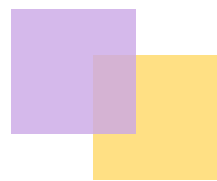
3.3.1 In-Domain Metric Depth Estimation

Method	Higher is better $\uparrow$			Lower is better $\downarrow$		
	δ_1	δ_2	δ_3	AbsRel	RMSE	log10
AdaBins [3]	0.903	0.984	0.997	0.103	0.364	0.044
DPT [47]	0.904	0.988	0.998	0.110	0.357	0.045
P3Depth [44]	0.898	0.981	0.996	0.104	0.356	0.043
SwinV2-L [40]	0.949	0.994	0.999	0.083	0.287	0.035
AiT [42]	0.954	0.994	0.999	0.076	0.275	0.033
VPD [87]	<u>0.964</u>	<u>0.995</u>	<u>0.999</u>	<u>0.069</u>	<u>0.254</u>	<u>0.030</u>
ZoeDepth* [4]	0.951	0.994	0.999	0.077	0.282	0.033
Ours	0.984	0.998	1.000	0.056	0.206	0.024

Table 3. **Fine-tuning and evaluating on NYUv2 [55]** with our pre-trained MDE encoder. We highlight **best**, second best results, as well as **most discriminative metrics**. *: Reproduced by us.

Method	Higher is better $\uparrow$			Lower is better $\downarrow$		
	δ_1	δ_2	δ_3	AbsRel	RMSE	RMSE log
AdaBins [3]	0.964	0.995	0.999	0.058	2.360	0.088
DPT [47]	0.959	0.995	0.999	0.062	2.573	0.092
P3Depth [44]	0.953	0.993	0.998	0.071	2.842	0.103
NeWCRFs [83]	0.974	0.997	0.999	0.052	2.129	0.079
SwinV2-L [40]	0.977	0.998	<u>1.000</u>	0.050	<u>1.966</u>	<u>0.075</u>
NDDepth [54]	<u>0.978</u>	<u>0.998</u>	0.999	0.050	2.025	0.075
GEDepth [76]	0.976	0.997	0.999	<u>0.048</u>	2.044	0.076
ZoeDepth* [4]	0.971	0.996	0.999	0.054	2.281	0.082
Ours	0.982	0.998	1.000	0.046	1.896	0.069

Table 4. **Fine-tuning and evaluating on KITTI [18]** with our pre-trained MDE encoder. *: Reproduced by us.



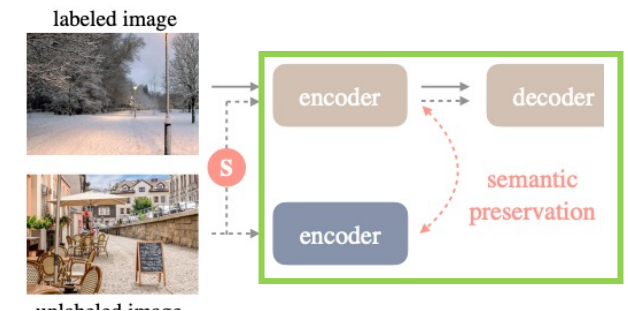
3.3.2 Zero-Shot Metric Depth Estimation

Method	SUN RGB-D [57]		iBims-1 [29]		HyperSim [49]		Virtual KITTI 2 [8]		DIODE Outdoor [60]	
	AbsRel ($\downarrow$)	δ_1 ($\uparrow$)	AbsRel	δ_1	AbsRel	δ_1	AbsRel	δ_1	AbsRel	δ_1
ZoeDepth [4]	0.520	0.545	0.169	0.656	0.407	0.302	0.106	0.844	0.814	0.237
Depth Anything	0.500	0.660	0.150	0.714	0.363	0.361	0.085	0.913	0.794	0.288

Table 5. **Zero-shot metric** depth estimation. The first three test sets in the header are indoor scenes, while the last two are outdoor scenes. Following ZoeDepth, we use the model trained on NYUv2 for indoor generalization, while use the model trained on KITTI for outdoor evaluation. For fair comparisons, we report the ZoeDepth results reproduced in our environment.

3.4 Fine-tuned to Semantic Segmentation

- In our method, we design our MDE model to inherit the rich semantic priors from a pre-trained encoder via a simple feature alignment constraint. Here, we examine the semantic capability of our MDE encoder.
- Specifically, we **fine-tune our MDE encoder to downstream semantic segmentation datasets**.



Method	Encoder	mIoU (s.s.)	m.s.
Segmenter [58]	ViT-L [16]	-	82.2
SegFormer [70]	MiT-B5 [70]	82.4	84.0
Mask2Former [12]	Swin-L [39]	83.3	84.3
OneFormer [24]	Swin-L [39]	83.0	84.4
OneFormer [24]	ConvNeXt-XL [41]	83.6	84.6
DDP [25]	ConvNeXt-L [41]	83.2	83.9
Ours	ViT-L [16]	84.8	86.2

Table 7. Transferring our MDE pre-trained encoder to **Cityscapes** for semantic segmentation. We **do not** use Mapillary [1] for pre-training. s.s./m.s.: single-/multi-scale evaluation.

Method	Encoder	mIoU
Segmenter [58]	ViT-L [16]	51.8
SegFormer [70]	MiT-B5 [70]	51.0
Mask2Former [12]	Swin-L [39]	56.4
UperNet [69]	BEiT-L [2]	56.3
ViT-Adapter [11]	BEiT-L [2]	58.3
OneFormer [24]	Swin-L [39]	57.4
OneFormer [24]	ConNeXt-XL [41]	57.4
Ours	ViT-L [16]	59.4

Table 8. Transferring our MDE encoder to **ADE20K** for semantic segmentation. We use Mask2Former as our segmentation model.

3.5 Qualitative Results

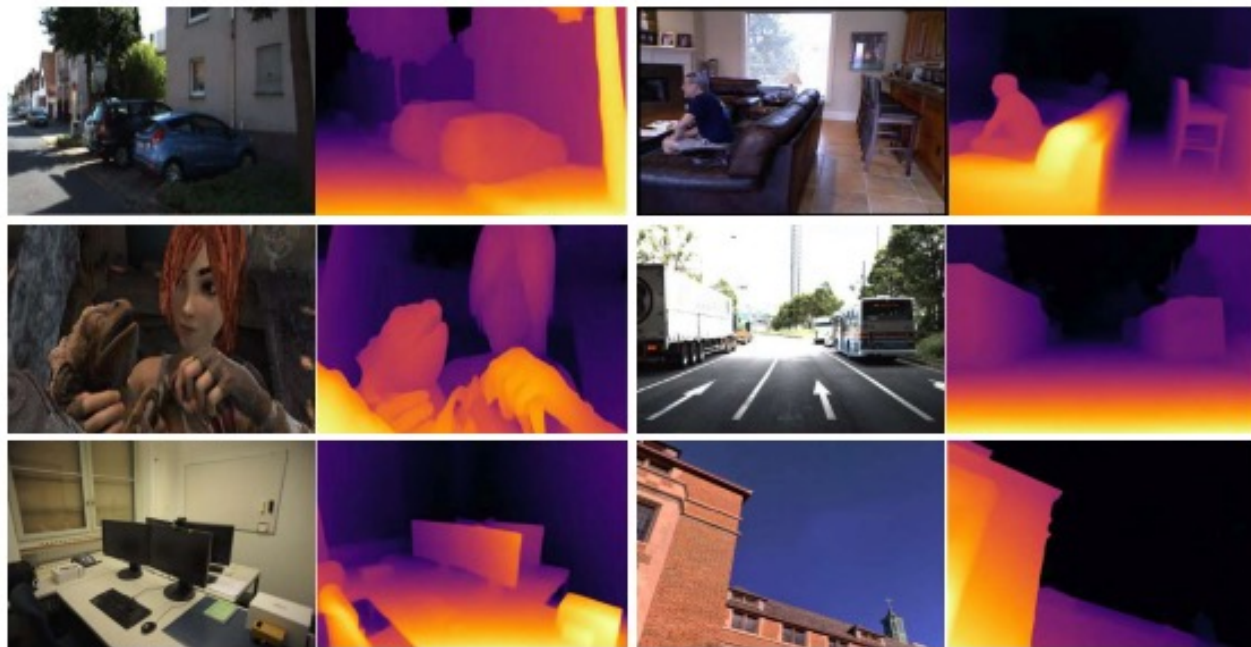


Figure 3. Qualitative results on six unseen datasets.

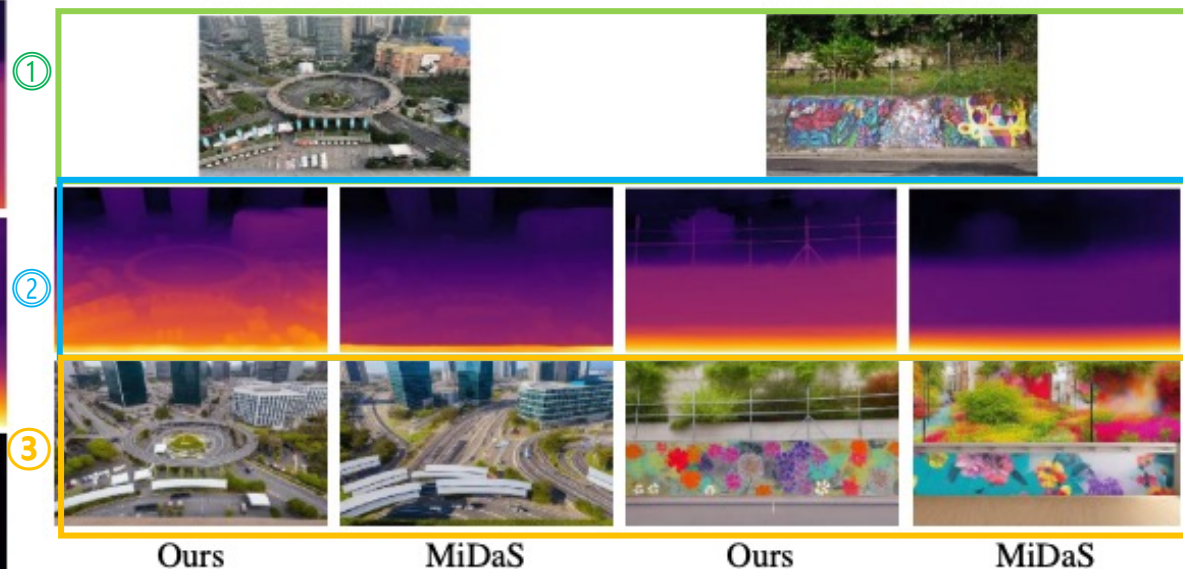
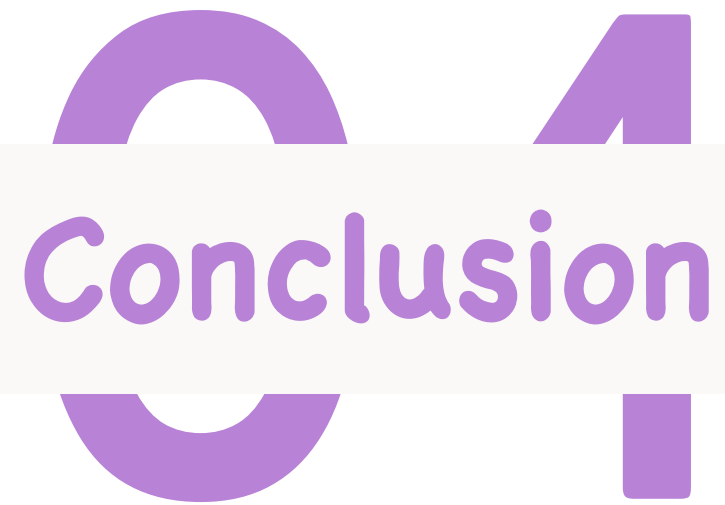


Figure 4. We compare our depth prediction with MiDaS. Meantime, we use ControlNet to synthesize new images from the depth map.



Conclusion



4 Conclusion

- In this work, we present Depth Anything, a highly practical solution to robust monocular depth estimation.
- Different from prior arts, we especially highlight the value of cheap and diverse unlabeled images.
- We design two simple yet highly effective strategies to fully exploit their value:
 - 1) posing a more challenging optimization target when learning unlabeled images
 - 2) preserving rich semantic priors from pre-trained models
- As a result, our Depth Anything model exhibits **excellent zero-shot depth estimation ability**, and also serves as **a promising initialization for downstream metric depth estimation and semantic segmentation tasks**.

Thank you for Listening!